2023

Improving Processing Speeds up to 50% Using AWS Graviton Instances with HERE Technologies

Learn how HERE Technologies, a location intelligence technology company, increased sustainability and performance using AWS Graviton–based instances.

Overview | Opportunity | Architectural Diagram | Solution | Outcome | AWS Services Used

Up to 60% fewer

peak-hour occasions needing extra processing capacity

faster responses for synchronous workloads

Up to 30%

cost reduction per request for asynchronous workloads

Up to 55%

cost reduction using Amazon EC2 Spot Instances

19% overall cost reduction

for the HERE Delivery Platform

Overview

Location intelligence technology company HERE Technologies (HERE) offers its Tour Planning API to help businesses optimize delivery routes, increasing sustainability and cost efficiency. Route optimization is a computationally intensive workload, and HERE sought a cloud-based solution from Amazon Web Services (AWS). For Tour Planning, HERE used the breadth of compute instance types available through Amazon Elastic Compute Cloud (Amazon EC2), secure and resizable compute capacity for virtually any workload. Using AWS, the Tour Planning team keeps costs low and increases sustainability for the company and its customers.

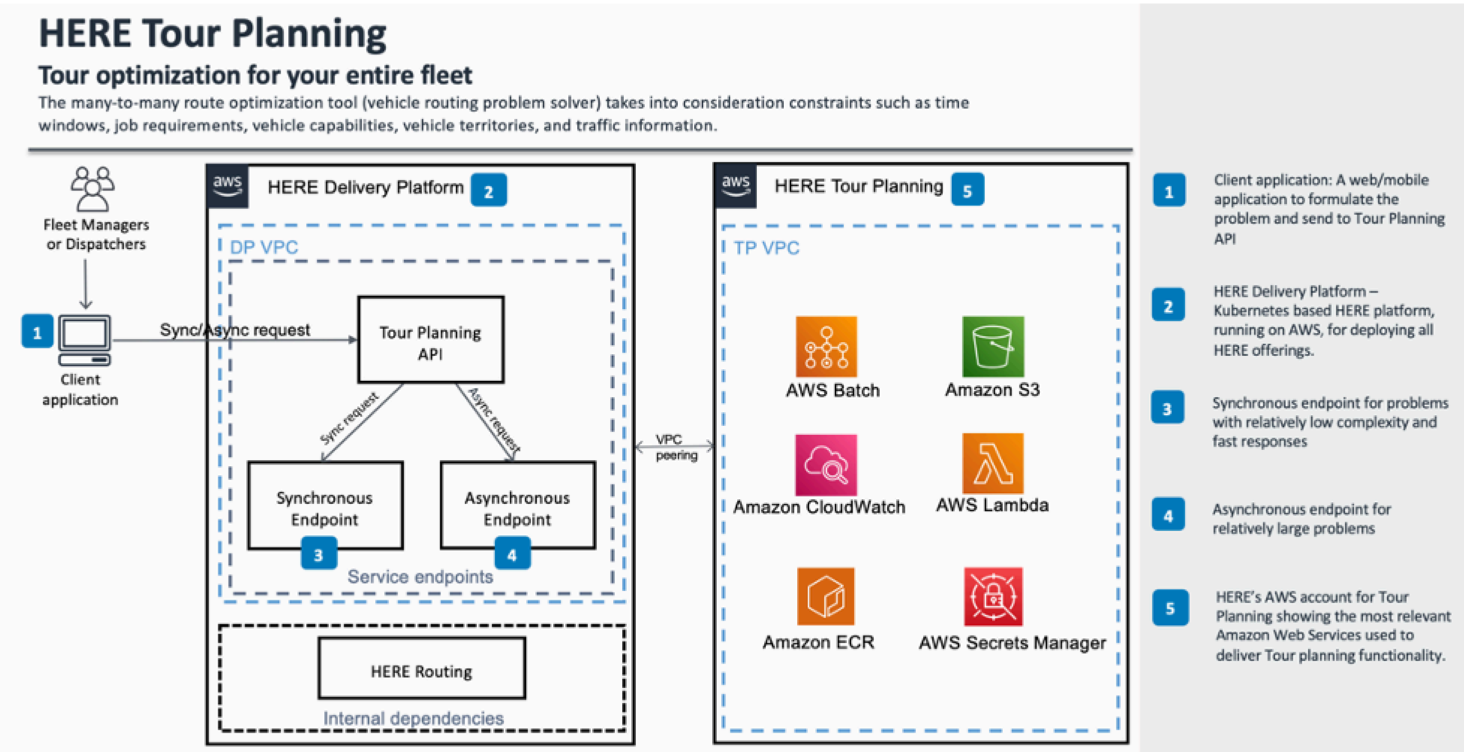
Opportunity | Reducing Costs and Improving Perform HERE's Tour Planning API Using AWS Graviton

With over 35 years of experience, HERE offers location technology to customers in more than 200 countries around the world. Its Tour Planning API is a versatile route planning and optimization tool that creates dynamic shift planning for last-mile fleets and minimizes cost or distance traveled based on available vehicles and drivers. It also maximizes throughput and deliveries at the lowest cost and is simple to use with existing in-house solutions and data. Customers can optimize routes based on parameters such as shortest time, shortest distance, or lowest cost. Tour Planning also uses a wide variety of additional route-optimization constraints, such as route restrictions based on near-real-time traffic data, to identify options for further cost and efficiency savings.

The final leg of a delivery, or "the last mile," from a local hub to the customer's home is considered the most expensive. Optimizing that route can greatly affect cost, customer experience, and environmental impact. To help customers with this challenge, HERE launched Tour Planning in 2020 using compute-optimized Amazon EC2 instances that run in Kubernetes clusters on the HERE Delivery Platform, a shared infrastructure with 18,000 deployments. Throughout the project, HERE received support from AWS representatives, meeting weekly to troubleshoot and strategize. With a focus on cost optimization, HERE participated in a deep-dive session in 2022 on AWS Batch, a fully managed batch computing service for running batch processing, machine learning model training, and analysis at nearly any scale. As part of the session, AWS introduced HERE to AWS Graviton processor, which delivers optimal price performance for cloud workloads running in Amazon EC2.

Architecture Diagram

Figure 1. HERE Tour Planning architecture diagram



Solution | Using AWS Graviton to Achieve up to 50 Percent Faster Responses and Increase Sustainability

In June 2022, the Tour Planning team began migrating its asynchronous infrastructure, which runs complex problems that take longer to yield results, to AWS Graviton–based instances. For 6 months, the team tested performance, availability, and cost impact in nonproduction workloads. Satisfied with the results, the team migrated its asynchronous production workloads to AWS Graviton–based instances in January 2023. Using AWS Batch and AWS Graviton–based instances for asynchronous workloads, HERE reduced costs by up to 30 percent per request while also increasing performance.

In January 2023, HERE's asynchronous workloads started using Amazon EC2 Spot Instances, which let users take advantage of unused Amazon EC2 capacity on AWS. HERE increased its use of Amazon EC2 Spot Instances on research and development deployments from 45 percent in 2022 to 88 percent by August 2023. This saved HERE up to 55 percent for its most common usage profiles. With these cost savings, HERE can offer the same service and performance to customers at a lower cost or use the savings to offer more features, capabilities, and performance to customers at the same price as before. "When we save on costs, we can be more competitive in terms of pricing for our customers," says Suresh Gupta, engineering manager for Tour Planning at HERE. "Or we can help them achieve even better results within the same runtime by factoring in more route-optimization variables."

After a lunch-and-learn session with a specialist for AWS Graviton–based instances in April 2023, the Tour Planning team began migrating its synchronous workloads—smaller, simpler problems that run in seconds and account for 90 percent of Tour Planning's API calls. Due to the robust testing HERE had done, the actual migration to AWS Graviton–based instances for the synchronous workloads took 1 week with zero downtime. "Usually, when we change part of our infrastructure, issues pop up, and we have to make fixes or adjustments later," says Tatu Niemi, lead DevOps engineer for the Tour Planning team at HERE. "Using AWS Graviton–based instances, we had no issues. It worked from the start."

HERE also improved processing speeds for synchronous workloads using AWS Graviton–based instances. "Fast response time matters more for synchronous workloads, and we saw up to 50 percent faster responses and sometimes even more," says Niemi. "For smaller optimization problems specifically, we see notably fast results." HERE also doesn't need to scale up as frequently or as high as before; it spins up larger-than-usual numbers of instances during peak hours 60 percent less often. HERE estimates that they are saving 10–15 percent for synchronous workloads because instances are less costly and the company needs to use them less frequently.

HERE aims to have net-zero carbon emissions by 2035. AWS Graviton–based instances use up to 60 percent less energy, which supports HERE's sustainability goals. "Using AWS services, HERE scales up elastic compute activity less often, using less energy and making our environmental footprint smaller," says Niemi. The company also increases sustainability by using fewer instances for Tour Planning. In addition, HERE helps its customers meet sustainability goals by minimizing distances traveled and avoiding traffic congestion, which helps to reduce carbon emissions.

Outcome | Continuing to Optimize Performance and Move Workloads to AWS Graviton

By adopting AWS Graviton–based instances for the HERE Delivery Platform, HERE has reduced costs by 19 percent overall. The Tour Planning team plans to continue optimizing performance for its AWS Graviton–based instances to further reduce costs, enhance the customer experience, and increase sustainability. In July 2023, the Tour Planning team started using Amazon Corretto—a no-cost, multiplatform, production-ready distribution of the Open Java Development Kit—to increase performance.

The Tour Planning team has migrated virtually 100 percent of its workload to AWS Graviton–based instances, setting an example for other HERE applications. As of August 2023, 30 percent of the workload on the HERE Delivery Platform uses AWS Graviton–based instances, and the company has the goal of boosting that number to 50 percent by the end of 2023. "Using AWS Graviton–based instances, we gain high-performance instances and reduce costs, which gives us the bandwidth to expand and support more complex use cases," says Gupta.

About HERE Technologies

HERE Technologies provides location intelligence technology to customers around the world. Its Tour Planning API offers customizable route optimization to help customers reduce costs and increase sustainability.

AWS Services Used

Amazon EC2 Spot Instances

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud and are available at up to a 90% discount compared to On-Demand prices.

Learn more »

AWS Graviton Processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

Learn more »

Amazon Corretto

Amazon Corretto is a no-cost, multiplatform, production-ready distribution of the Open Java Development Kit (OpenJDK).

Learn more »

AWS Batch

AWS Batch lets developers, scientists, and engineers efficiently run hundreds of thousands of batch and ML computing jobs while optimizing compute resources, so you can focus on analyzing results and solving problems.

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From one of the most recognized tire brands in the world to a connected mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to the cloud and provides a virtual

2024

2024

ITALY

Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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